

Task on CloudTrail & CloudWatch

1. **Enable CloudTrail monitoring and store the events in S3 and CloudWatch log events.**

Implementation

1. Log in to the **AWS Management Console**.
2. Open **CloudTrail** → **Trails**.
3. Click **Create trail**.
4. Enter a trail name, for example devops-cloudtrail.
5. Select **Create new S3 bucket** for log storage.
6. Enter a unique S3 bucket name.
7. Enable **Management events**.
8. Select **Read and Write events**.
9. Enable **CloudWatch Logs**.
10. Enter a CloudWatch Log Group name, for example /aws/cloudtrail/devops.
11. Select or create the required **IAM role** for CloudWatch Logs.
12. Click **Create trail**.
13. Verify that the trail status is **Logging**.
14. Perform an AWS activity such as starting or stopping an EC2 instance.
15. Check **CloudTrail** → **Event history**.
16. Verify the CloudTrail logs in the **S3 bucket**.
17. Open **CloudWatch** → **Log groups**.
18. Open the CloudTrail log group.
19. Verify that CloudTrail events are available.

us-east-2.console.aws.amazon.com/cloudtrailv2/home?region=us-east-2#/create

CloudTrail > Trails > Create trail

Step 3
Review and create

Trail name
Enter a display name for your trail.

3-128 characters. Only letters, numbers, periods, underscores, and dashes are allowed.

☐ Enable for all accounts in my organization
To review accounts in your organization, open AWS Organizations. [See all accounts.](#)

Storage location [Info](#)
☒ Create new S3 bucket
Create a bucket to store logs for the trail.
☐ Use existing S3 bucket
Choose an existing bucket to store logs for this trail.

Trail log bucket and folder
Enter a new S3 bucket name and folder (prefix) to store your logs. Bucket names must be globally unique.

Logs will be stored in aws-cloudtrail-logs-281712468655-56b5601c/AWSLogs/281712468655

Log file SSE-KMS encryption [Info](#)
☐ Enabled

us-east-2.console.aws.amazon.com/cloudtrailv2/home?region=us-east-2#/create

CloudTrail > Trails > Create trail

Configure CloudWatch Logs to monitor your trail logs and notify you when specific activity occurs. Standard CloudWatch and CloudWatch Logs charges apply. [Learn more](#)

CloudWatch Logs [Info](#)
☒ Enabled

Log group [Info](#)
☒ New
☐ Existing

Log group name

1-512 characters. Only letters, numbers, dashes, underscores, forward slashes, and periods are allowed.

IAM Role [Info](#)
AWS CloudTrail assumes this role to send CloudTrail events to your CloudWatch Logs log group.
☒ New
☐ Existing

Role name

us-east-2.console.aws.amazon.com/cloudtrailv2/home?region=us-east-2#/create

CloudTrail > Trails > Create trail

Step 2
Choose log events

Events [Info](#)
Record API activity for individual resources, or for all current and future resources in AWS account. [Additional charges apply](#)

Event type
Choose the type of events that you want to log.
☒ Management events
Capture management operations performed on your AWS resources.
☐ Data events
Log the resource operations performed on or within a resource.
☐ Insights events
Identify unusual activity, errors, or user behavior in your account.
☐ Network activity events
Network activity events provide information about resource operations performed on a resource within a virtual private cloud endpoint.

Management events [Info](#)
Management events show information about management operations performed on resources in your AWS account.

us-east-2.console.aws.amazon.com/cloudtrailv2/home?region=us-east-2#/create

CloudTrail > Trails > Create trail

Step 2
Choose log events

Step 3
Review and create

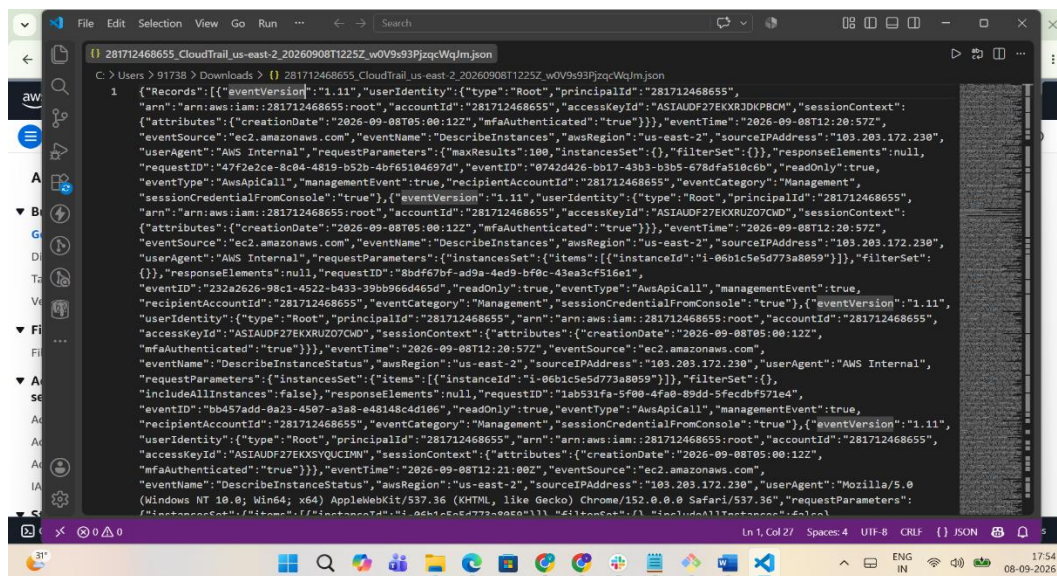
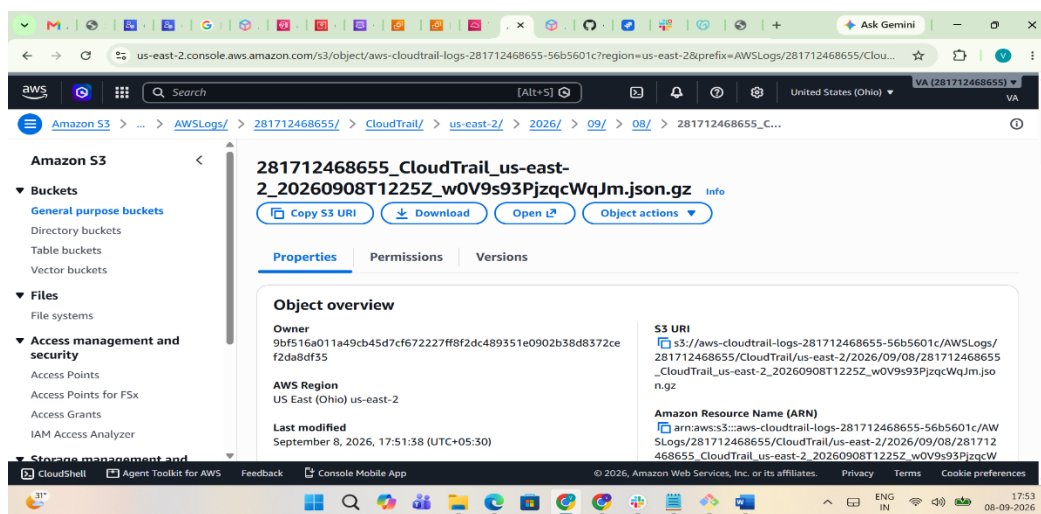
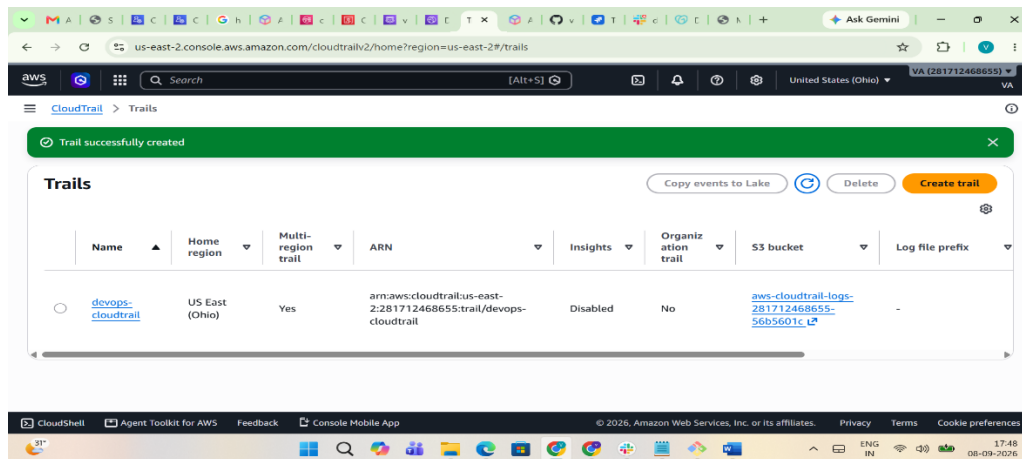
Step 1: Choose trail attributes [Edit](#)

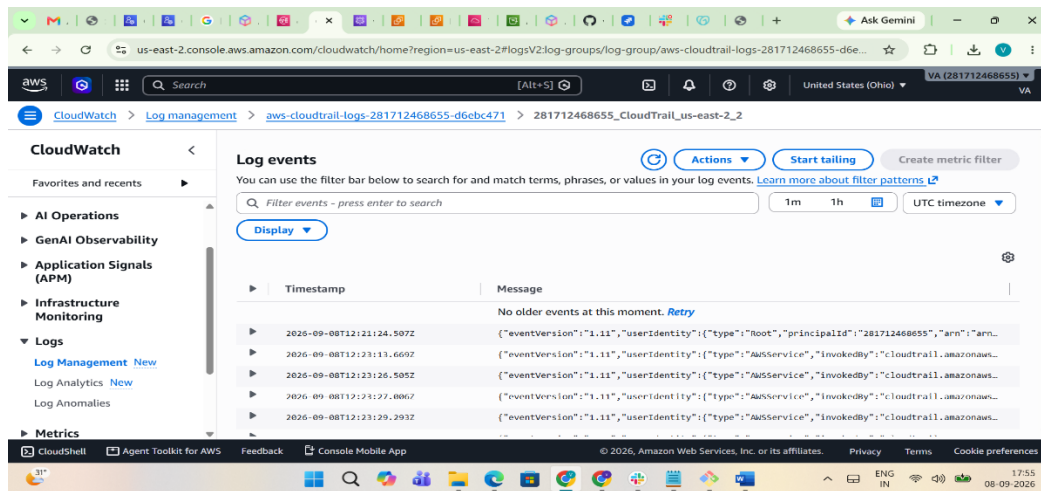
General details

Trail name devops-cloudtrail	Trail log location aws-cloudtrail-logs-281712468655-56b5601c/AWSLogs/281712468655	Log file validation Enabled
Multi-region trail Yes	Log file SSE-KMS encryption Not enabled	Recursive logging Enabled
Apply trail to my organization Not enabled		SNS notification delivery arn:aws:sns:us-east-2:281712468655:cloudtrail-alerts

CloudWatch Logs

Log group aws-cloudtrail-logs-281712468655-d6ebc471	IAM Role CloudTrailRoleForCloudwatchlogs_devops-cloudtrail
---	--

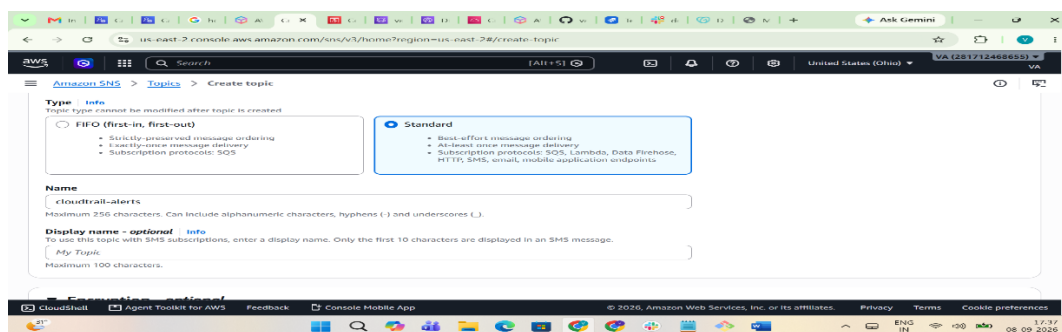


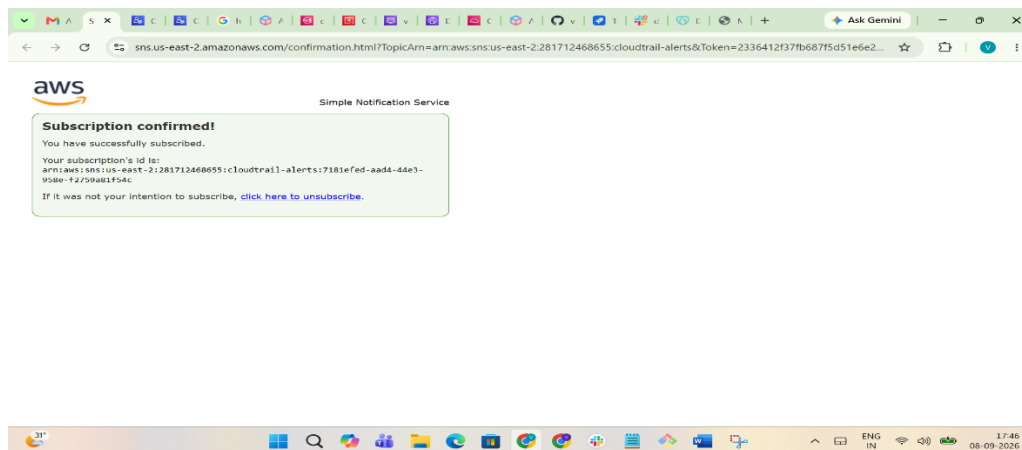
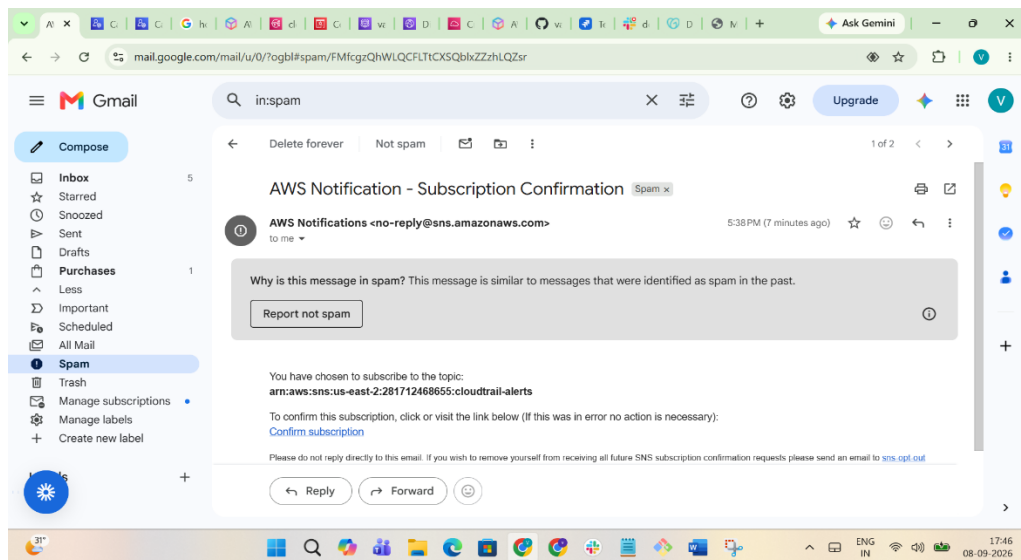
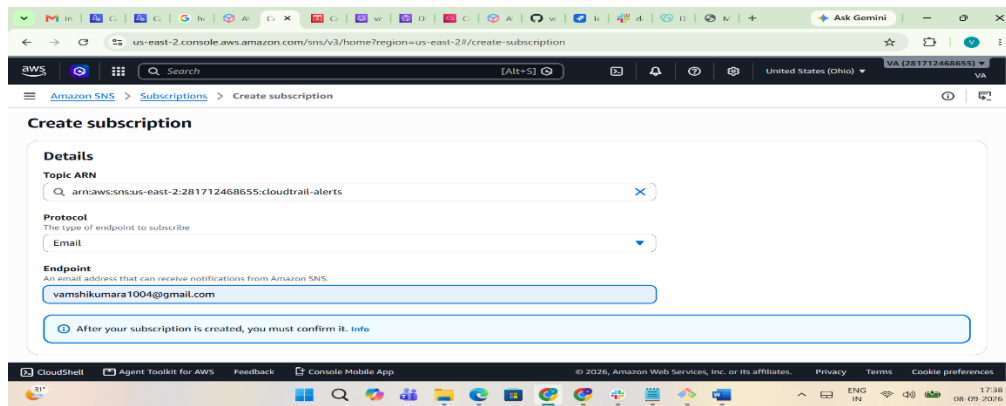


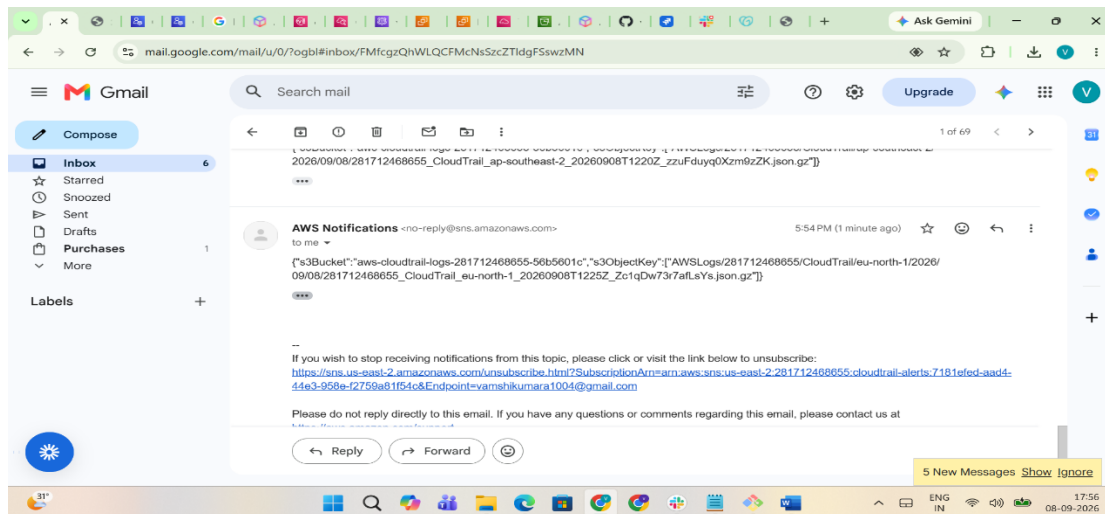
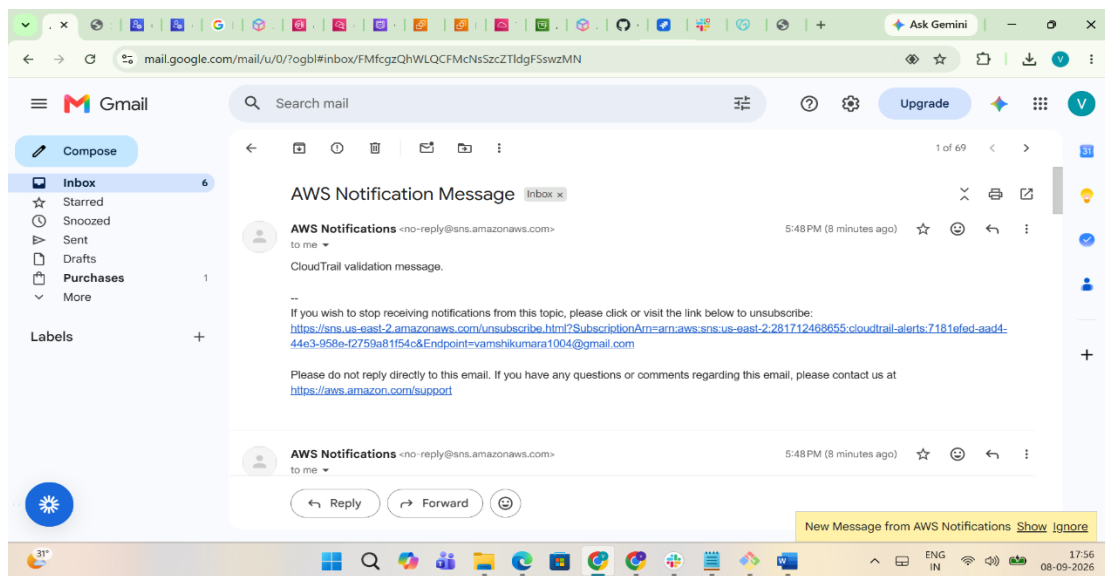
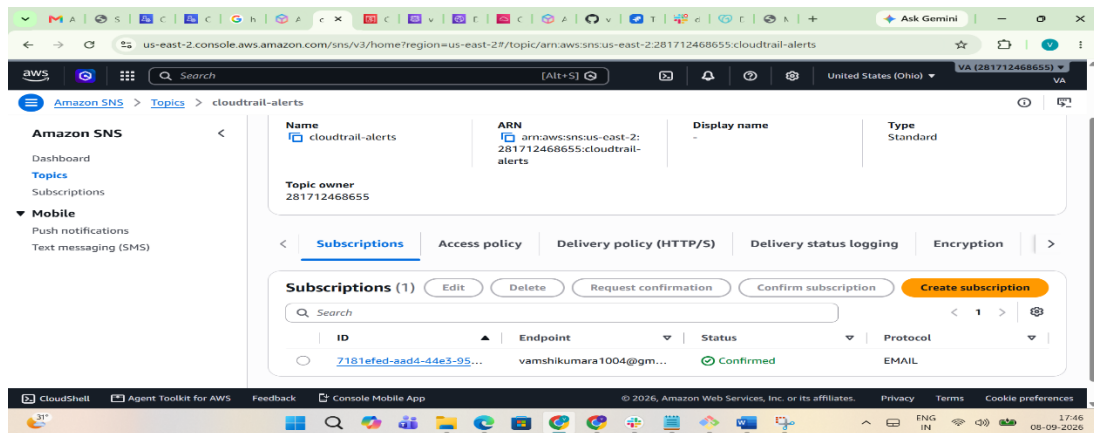
2. Enable SNS for CloudTrail to send alerts via email.

Implementation

1. Log in to the **AWS Management Console**.
2. Open **Amazon SNS** → **Topics**.
3. Click **Create topic**.
4. Select **Standard**.
5. Enter a topic name, for example cloudtrail-alerts.
6. Click **Create topic**.
7. Open the created SNS topic.
8. Click **Create subscription**.
9. Select **Email** as the protocol.
10. Enter the required email address.
11. Click **Create subscription**.
12. Open the confirmation email received from AWS.
13. Click **Confirm subscription**.
14. Verify that the subscription status is **Confirmed**.
15. Configure CloudTrail/CloudWatch to use this SNS topic for alerts.
16. Perform the configured CloudTrail activity.
17. Verify that the alert is received in the email.





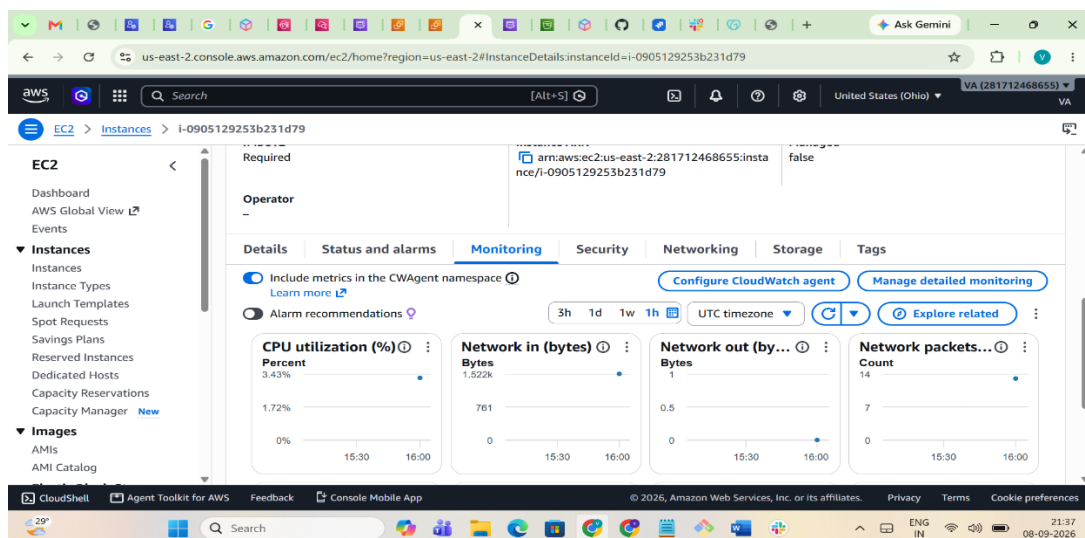


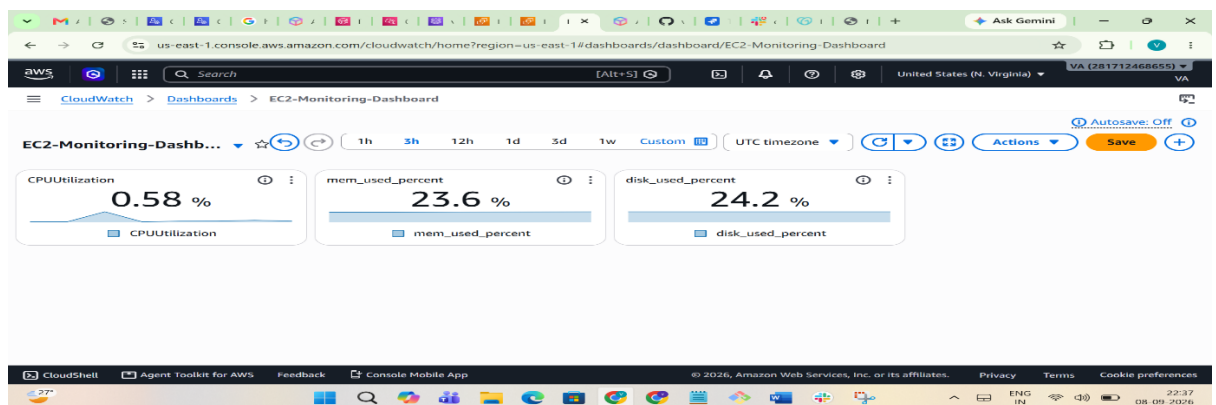
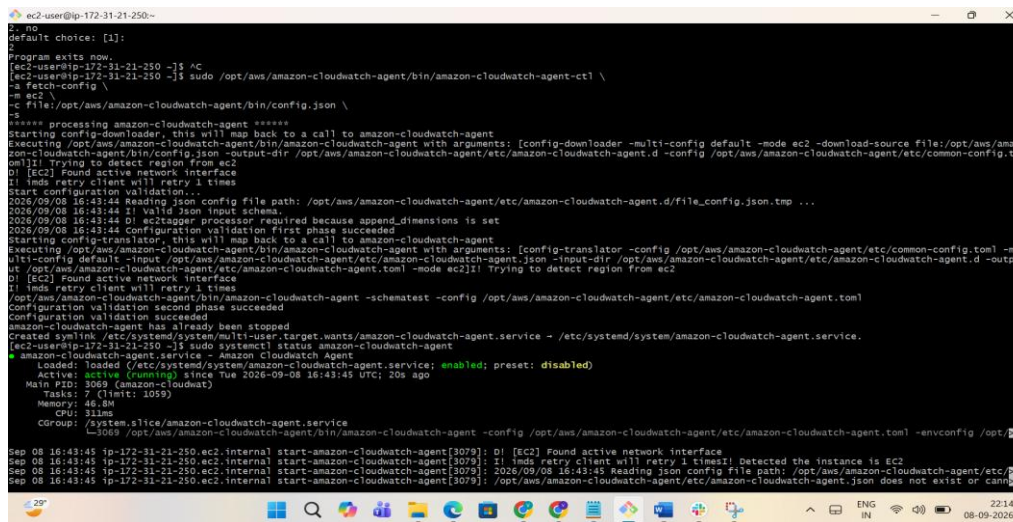
3. Configure CloudWatch monitoring and record the CPU utilization and other metrics of EC2.

Implementation

1. Log in to the AWS Management Console.

2. Open **EC2** → **Instances**.
3. Select the required EC2 instance.
4. Open the **Monitoring** tab.
5. Verify the default EC2 metrics.
6. Open **CloudWatch** → **Metrics**.
7. Select **EC2** → **Per-Instance Metrics**.
8. Select the required EC2 instance.
9. Select **CPUUtilization**.
10. Verify the CPU utilization graph.
11. Configure the **CloudWatch Agent** on the EC2 instance for additional metrics.
12. Install the CloudWatch Agent.
13. Attach the required IAM role to the EC2 instance.
14. Run the CloudWatch Agent configuration wizard.
15. Configure **CPU, memory, disk, and network metrics** as required.
16. Start the CloudWatch Agent.
17. Open **CloudWatch** → **Metrics** → **CWAgent**.
18. Verify that the additional EC2 metrics are being recorded.

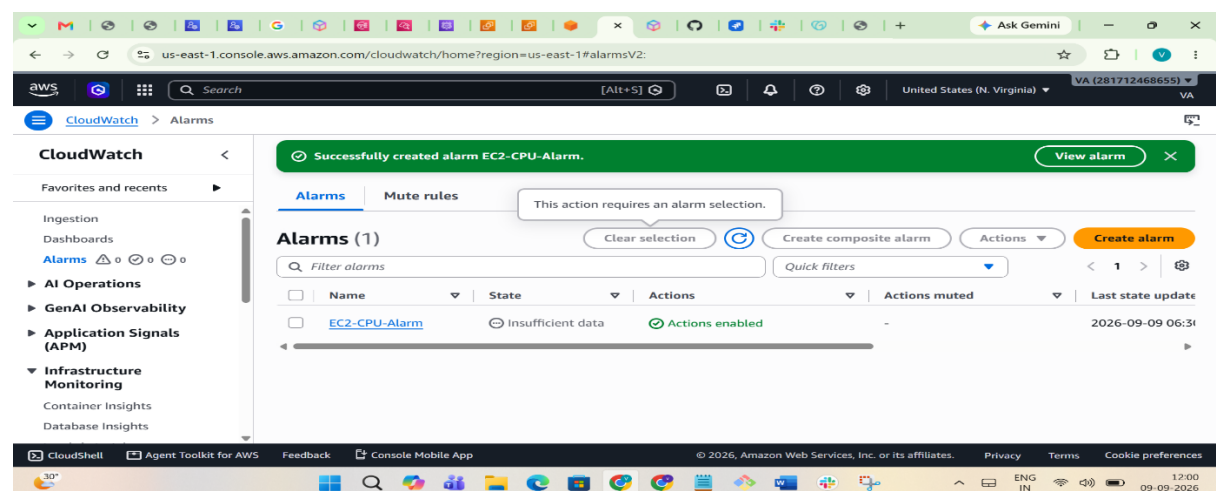


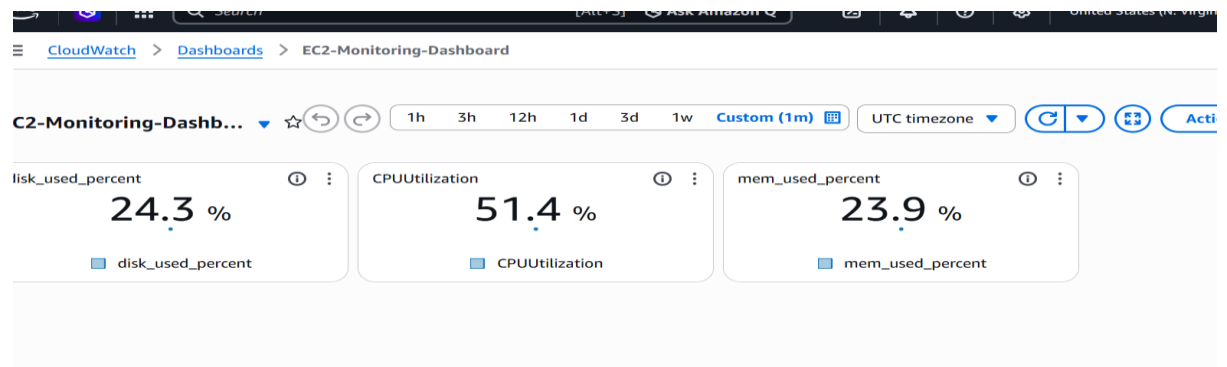


4. Create one alarm to send an alert to email if the CPU utilization is more than 70 percent.

Implementation

1. Open **CloudWatch** → **Alarms**.
2. Click **Create alarm**.
3. Click **Select metric**.
4. Select **EC2** → **Per-Instance Metrics**.
5. Select the required EC2 instance.
6. Select **CPUUtilization**.
7. Select the required statistic, such as **Average**.
8. Select a monitoring period, such as **5 minutes**.
9. Set the threshold condition to **Greater than 70**.
10. Select the **In alarm** action.
11. Select the previously created **SNS topic**.
12. Enter an alarm name, for example **EC2-CPU-Above-70**.
13. Click **Create alarm**.
14. Generate CPU load on the EC2 instance for testing.
15. Verify that CPU utilization exceeds **70%**.
16. Verify that the CloudWatch alarm changes to **ALARM**.
17. Verify that an email alert is received through SNS.
18. Stop the test CPU load after verification.





Alarms

Mute rules

Alarms (1)

Clear selection

Create composite alarm

Actions

▼

Create alarm

Filter alarms

Quick filters

▼

<

1

>

☐	Name	State	Actions	Actions muted	Last state update
☐	EC2-CPU-Alarm	OK	Actions enabled	-	2026-09-09 06:31

The screenshot displays the AWS Management Console interface for an EC2 instance. The browser address bar shows the URL: `us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#InstanceDetails:instanceId=i-055850127ddec40f1`. The console header includes the AWS logo, a search bar, and navigation links for EC2, Instances, and the specific instance ID `i-055850127ddec40f1`. The left-hand navigation pane lists various EC2-related options, with 'Instances' selected. The main content area is titled 'Instance summary for i-055850127ddec40f1 (ec2-vir-sever)' and includes an 'Info' link. Below the title are buttons for 'Connect', 'Instance state', and 'Actions'. The summary is organized into a grid of key-value pairs:

- Instance ID:** i-055850127ddec40f1
- IPV6 address:** -
- Hostname type:** IP name: ip-172-31-21-250.ec2.internal
- Answer private resource DNS name:** IPV4 (A)
- Auto-assigned IP address:** 13.218.89.10 [Public IP]
- Public IPV4 address:** 13.218.89.10 | [open address](#)
- Instance state:** Running
- Private IP DNS name (IPV4 only):** ip-172-31-21-250.ec2.internal
- Instance type:** t3.micro
- VPC ID:** vpc-0c90519b46e7b972f (vpc-virgina) | [open vpc](#)
- Private IPV4 addresses:** 172.31.21.250
- Public DNS:** ec2-13-218-89-10.compute-1.amazonaws.com | [open address](#)
- Elastic IP addresses:** -
- AWS Compute Optimizer finding:** Opt-in to AWS Compute Optimizer for recommendations.

The bottom of the console shows a footer with links for CloudShell, Agent Toolkit for AWS, Feedback, Console Mobile App, and a copyright notice for Amazon Web Services, Inc. The system tray at the very bottom indicates the date and time as 09-09-2026, 12:25.

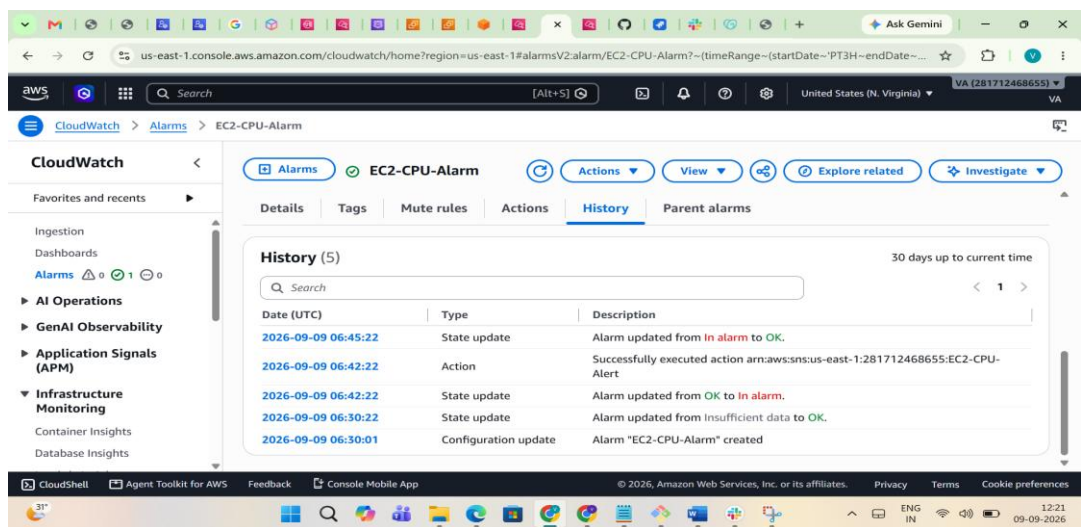
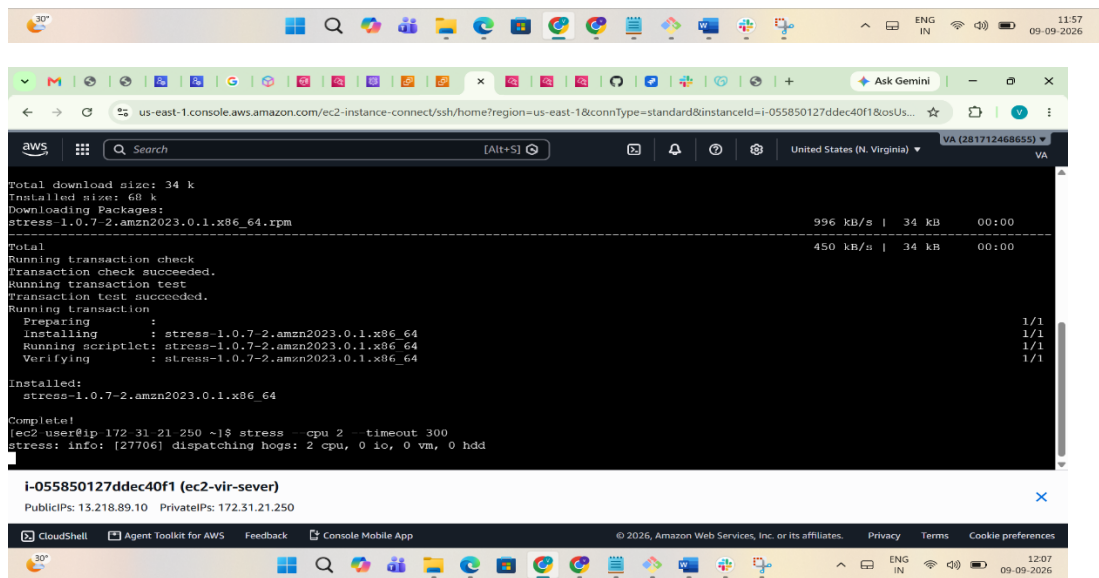
The screenshot displays the Amazon SNS console for the topic 'EC2-CPU-Alert'. The console shows the following details:

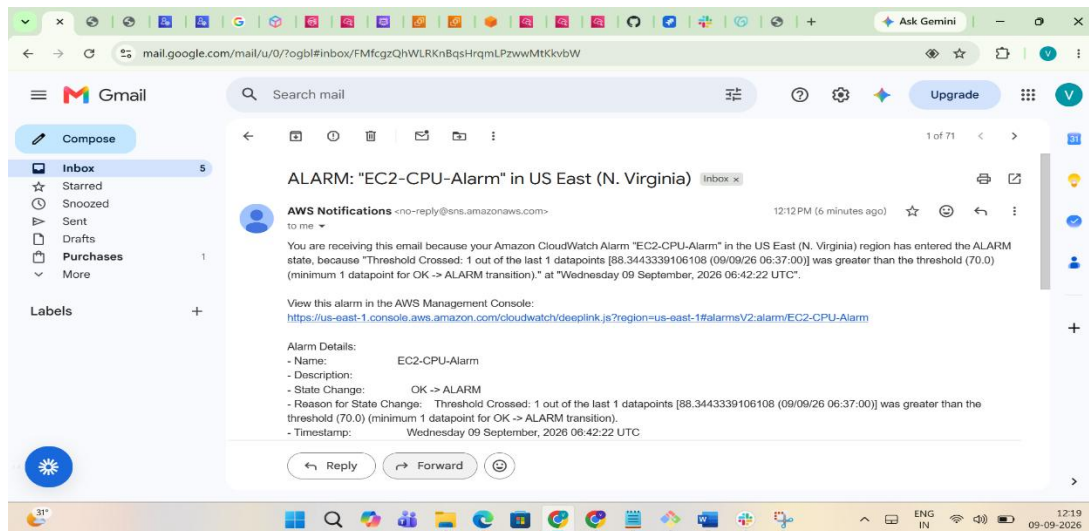
- Name:** EC2-CPU-Alert
- ARN:** arn:aws:sns:us-east-2:281712468655:EC2-CPU-Alert
- Display name:** -
- Type:** Standard
- Topic owner:** 281712468655

The 'Subscriptions' tab is active, showing a list of subscriptions. The table has the following columns: ID, Endpoint, Status, Protocol, and a search bar. The table contains one subscription:

ID	Endpoint	Status	Protocol
16e369a0-7263-41b3-8...	vamshikumara1004@gmail...	Confirmed	EMAIL

The bottom of the screenshot shows the AWS CloudShell and Agent Toolkit for AWS.





5. Create a Dashboard and monitor the Tomcat service whether it is running or not and send the alert.

Implementation

1. Log in to the **EC2 Linux instance**.
2. Verify the java –version then install tomcat
3. Verify the Tomcat service: wget <https://d1cdn.apache.org/tomcat/tomcat-9/v9.0.121/bin/apache-tomcat-9.0.121.tar.gz> and then ls
4. Extract the tar file and run the tomcat
5. Configure the script to publish **Tomcat status** as a custom CloudWatch metric.

Ex: monitoring.bash

```
#!/bin/bash
```

```
#set -x
```

```
# Get EC2 Instance ID using IMDSv2
```

```
TOKEN=$(curl -s -X PUT \
"http://169.254.169.254/latest/api/token" \
-H "X-aws-ec2-metadata-token-ttl-seconds: 21600")
```

```
INSTANCE_ID=$(curl -s \
-H "X-aws-ec2-metadata-token: $TOKEN" \
"http://169.254.169.254/latest/meta-data/instance-id")
```

```

checkTomcatStatus(){
    counter=0

    ps x | grep /opt/apache-tomcat-9.0.121/ | grep -v grep | #Need to change
tomcat version

    while read -r LINE
    do

        read PROCESS_ID <<< $LINE

        counter=$((counter+1))

        echo $counter

    done
}

i=$(checkTomcatStatus)

#echo $i

if [ "$i" == "" ]
then

    aws --region us-east-1 cloudwatch put-metric-data --metric-name tomcat --value
0 --namespace tomcat --dimensions InstanceId=$INSTANCE_ID

else

    aws --region us-east-1 cloudwatch put-metric-data --metric-name tomcat --value
1 --namespace tomcat --dimensions InstanceId=$INSTANCE_ID

fi

```

Need to change the region

6. Configure: 1 = Tomcat running 0 = Tomcat stopped
7. Give the EC2 IAM role permission to publish CloudWatch metrics.

Ex: CustomCloudWatchMetricPolicy

```
{
```

```
"Version": "2012-10-17",  
"Statement": [  
  {  
    "Sid": "AllowPutCustomMetrics",  
    "Effect": "Allow",  
    "Action": [  
      "cloudwatch:PutMetricData"  
    ],  
    "Resource": "*"    
  }  
]
```

8. Schedule the monitoring script using **cron**.
9. Open **CloudWatch → Metrics**.
10. Verify the custom **TomcatStatus** metric.
11. Create a CloudWatch alarm for **TomcatStatus = 0**.
12. Configure the alarm to send notifications through the **SNS topic**.
13. Open **CloudWatch → Dashboards**.
14. Click **Create dashboard**.
15. Enter a dashboard name, for example Server-Monitoring.
16. Add the **TomcatStatus** metric to the dashboard.
17. Save the dashboard.
18. Stop Tomcat: `sudo systemctl stop tomcat`
19. Verify that the CloudWatch alarm changes to **ALARM**.
20. Verify that the email alert is received.
21. Start Tomcat again and verify that the alarm returns to **OK**.

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#InstanceDetails:instanceId=i-055850127ddec40f1

EC2 > Instances > i-055850127ddec40f1

Instance summary for i-055850127ddec40f1 (ec2-vir-sever) [Info](#)

Updated less than a minute ago

[Connect](#) [Instance state](#) [Actions](#)

Instance ID i-055850127ddec40f1	Public IPv4 address 13.218.89.10 open address	Private IPv4 addresses 172.31.21.250
IPv6 address -	Instance state Running	Public DNS ec2-13-218-89-10.compute-1.amazonaws.com open address
Hostname type IP name: ip-172-31-21-250.ec2.internal	Private IP DNS name (IPv4 only) ip-172-31-21-250.ec2.internal	Elastic IP addresses -
Answer private resource DNS name IPv4 (A)	Instance type t3.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations.
Auto-assigned IP address 13.218.89.10 [Public IP]	VPC ID vpc-0c90519b46e7b972f (vpc-virgina)	

CloudShell Agent Toolkit for AWS Feedback Console Mobile App

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```
51738@IDEAPAD MINGW64 ~/Downloads
$ ssh -i linux-testing.pem ec2-user@13.218.89.10
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Wed Sep  9 07:23:42 2026 from 103.203.172.230
[ec2-user@ip-172-31-21-250 ~]$ cd /pot
-bash: cd: /pot: No such file or directory
[ec2-user@ip-172-31-21-250 ~]$ cd /opt/
[ec2-user@ip-172-31-21-250 opt]$ ls
apache-tomcat-9.0.121  apache-tomcat-9.0.121.tar.gz  aws  monitoring.bash
[ec2-user@ip-172-31-21-250 opt]$ cd apache-tomcat-9.0.121/
[ec2-user@ip-172-31-21-250 apache-tomcat-9.0.121]$ ls
BUILDING.txt  CONTRIBUTING.md  LICENSE  NOTICE  README.md  RELEASE-NOTES  RUNNING.txt  bin  conf  lib  logs  temp  webapps  work
Using CATALINA_BASE:   /opt/apache-tomcat-9.0.121
Using CATALINA_HOME:   /opt/apache-tomcat-9.0.121
Using CATALINA_TMPDIR: /opt/apache-tomcat-9.0.121/temp
Using JRE_HOME:        /usr
Using CLASSPATH:        /opt/apache-tomcat-9.0.121/bin/bootstrap.jar:/opt/apache-tomcat-9.0.121/bin/tomcat-juli.jar
Tomcat started.
```

us-east-1.console.aws.amazon.com/iam/home?region=us-east-2#/roles/details/EC2-CloudWatchAgent-Role?section=permissions

IAM > Roles > EC2-CloudWatchAgent-Role

Identity and Access Management (IAM)

Search IAM

Dashboard

Access Management

Roles

Policies

IAM users

IAM user groups

Identity providers

Account settings

Root access management

Temporary delegation requests

Access reports

CloudShell Agent Toolkit for AWS Feedback Console Mobile App

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us-east-1.console.aws.amazon.com/cloudwatch/home?region=us-east-1#alarmsV2:create?~(Page=~MetricSelection~metricType=~classic~AlarmType=~...

CloudWatch > Alarms > Create alarm

Metric

Namespace: tomcat

Metric name: tomcat

InstanceId: i-055850127ddec40f1

Instance name: ec2-vir-sever

Statistic: Minimum

Period: 1 minute

Conditions

Threshold type: Static

Whenever tomcat is...: Greater > threshold

Preview

1h 3h 12h 1d 3d 1w Custom UTC timezone

Graph

This alarm will trigger when the blue line goes below the red line for 1 datapoints within 1 minute.

us-east-1.console.aws.amazon.com/cloudwatch/home?region=us-east-1#dashboards/dashboard/EC2-Monitoring-Dashboard?start=PT1M&end=null

CloudWatch > Dashboards > EC2-Monitoring-Dashboard

EC2-Monitoring-Dashb...

tomcat 1

tomcat

sns.us-east-1.amazonaws.com/confirmation.html?TopicArn=arn:aws:sns:us-east-1:281712468655:EC2-Service-Alert&Token=2336412f37fb687f5d51e6e...

aws Simple Notification Service

Subscription confirmed!

You have successfully subscribed.

Your subscription's Id is: `arn:aws:sns:us-east-1:281712468655:EC2-Service-Alert:a2fe8bde-9ed3-4c18-ab43-fd0a06ecfdcf`

If it was not your intention to subscribe, [click here to unsubscribe](#).

```
[ec2-user@ip-172-31-21-250 opt]$ sudo /opt/apache-tomcat-9.0.121/bin/shutdown.sh
Using CATALINA_BASE:   /opt/apache-tomcat-9.0.121
Using CATALINA_HOME:   /opt/apache-tomcat-9.0.121
Using CATALINA_TMPDIR: /opt/apache-tomcat-9.0.121/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/apache-tomcat-9.0.121/bin/bootstrap.jar:/opt/apache-tomcat-9.0.121/bin/tomcat-juli.jar
Using CATALINA_OPTS:
NOTE: Picked up JDK_JAVA_OPTIONS:  --add-opens=java.base/java.lang=ALL-UNNAMED --add-opens=java.base/java.lang.invoke=ALL-UNNAMED --add-opens=java.base/java.lang.reflect=ALL-UNNAMED --add-opens=java.base/java.io=ALL-UNNAMED --add-opens=java.base/java.util=ALL-UNNAMED --add-opens=java.base/java.util.concurrent=ALL-UNNAMED --add-opens=java.rmi/sun.rmi.transport=ALL-UNNAMED
```

us-east-1.console.aws.amazon.com/sns/v3/home?region=us-east-1#/topic/am:aws:sns:us-east-1:281712468655:EC2-Service-Alert

Amazon SNS > Topics > EC2-Service-Alert

Amazon SNS

- Dashboard
- Topics
- Subscriptions
- Mobile
 - Push notifications
 - Text messaging (SMS)

Details

Name EC2-Service-Alert	ARN arn:aws:sns:us-east-1:281712468655:EC2-Servi ce-Alert	Display name -	Type Standard
Topic owner 281712468655			

Subscriptions (1) [Edit](#) [Delete](#) [Request confirmation](#) [Confirm subscription](#) [Create subscription](#)

ID	Endpoint	Status	Protocol
a2fe8d5e-9ed3-4c18-ab...	vamshikumara1004@gm...	Confirmed	EMAIL

us-east-1.console.aws.amazon.com/cloudwatch/home?region=us-east-1#dashboards/dashboard/EC2-Monitoring-Dashboard?start=PT1M&end=null

CloudWatch > Dashboards > EC2-Monitoring-Dashboard

EC2-Monitoring-Dashb... [Refresh](#) [Reset](#) 1h 3h 12h 1d 3d 1w Custom (1m) UTC timezone [Actions](#) [Save](#)

tomcat

0

tomcat

us-east-1.console.aws.amazon.com/cloudwatch/home?region=us-east-1#alarmsV2:alarm/ec2-service-alarm?~(timeRange~(startDate~PT3H~endDate...

CloudWatch > Alarms > ec2-service-alarm

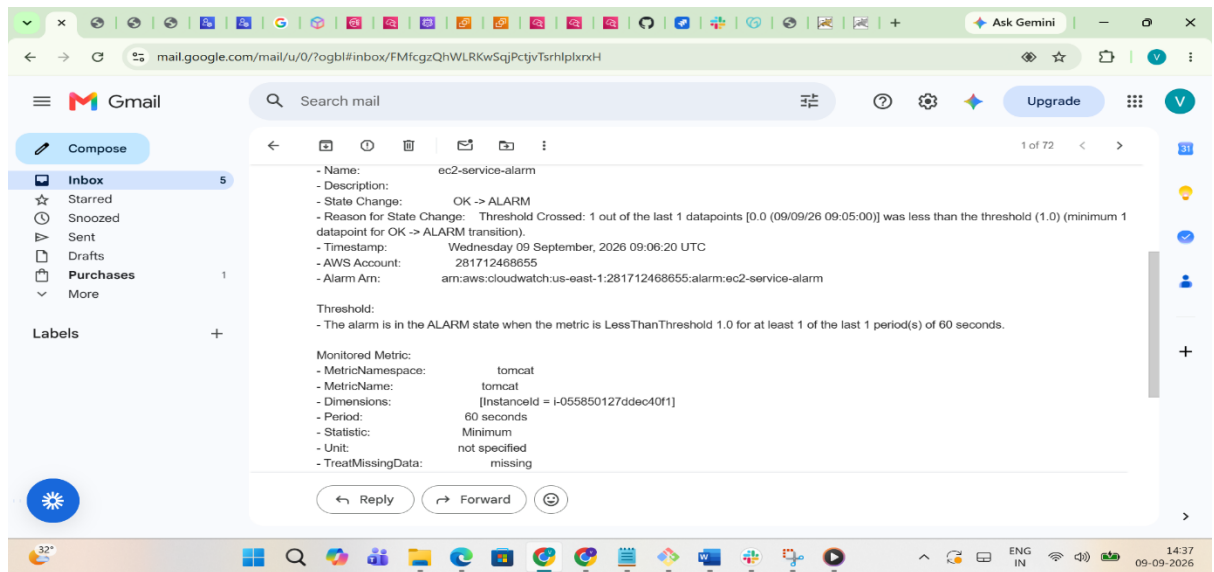
CloudWatch

- Favorites and recents
- Ingestion
- Dashboards
- Alarms 1 OK OK OK
- AI Operations
- GenAI Observability
- Application Signals (APM)
- Infrastructure Monitoring
 - Container Insights
 - Database Insights

Alarms ec2-service-alarm [Actions](#) [View](#) [Explore related](#) [Investigate](#)

History (4) 30 days up to current time

Date (UTC)	Type	Description
2026-09-09 09:06:20	Action	Successfully executed action arn:aws:sns:us-east-1:281712468655:EC2-Service-Alert
2026-09-09 09:06:20	State update	Alarm updated from OK to In alarm.
2026-09-09 08:58:20	State update	Alarm updated from Insufficient data to OK.
2026-09-09 08:56:57	Configuration update	Alarm "ec2-service-alarm" created



6. Create a Dashboard and monitor the Nginx service to send the alert if Nginx is not running.

Implementation

1. Log in to the **EC2 Linux instance**.
2. Verify the Nginx service: `sudo systemctl status nginx`
3. Create a script to check the Nginx service status.
4. Configure the script to publish **Nginx status** as a custom CloudWatch metric.

Ex: monitoring-nginx.bash

```
#!/bin/bash
```

```
# Get EC2 Instance ID using IMDSv2
```

```
TOKEN=$(curl -s -X PUT \
"http://169.254.169.254/latest/api/token" \
-H "X-aws-ec2-metadata-token-ttl-seconds: 21600")
```

```
INSTANCE_ID=$(curl -s \
-H "X-aws-ec2-metadata-token: $TOKEN" \
"http://169.254.169.254/latest/meta-data/instance-id")
```

```
# Check Nginx service status
```

```

if systemctl is-active --quiet nginx
then
    aws --region us-east-1 cloudwatch put-metric-data \
        --metric-name nginx \
        --value 1 \
        --namespace nginx \
        --dimensions InstanceId="$INSTANCE_ID"

    echo "Nginx is running"
else
    aws --region us-east-1 cloudwatch put-metric-data \
        --metric-name nginx \
        --value 0 \
        --namespace nginx \
        --dimensions InstanceId="$INSTANCE_ID"

    echo "Nginx is NOT running"
fi

```

5. Configure: 1 = Nginx running 0 = Nginx stopped
6. Give the EC2 IAM role permission to publish CloudWatch metrics.

Ex: CustomCloudWatchMetricPolicy

```

{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AllowPutCustomMetrics",
      "Effect": "Allow",
      "Action": [
        "cloudwatch:PutMetricData"
      ]
    }
  ]
}

```

```
    ],  
    "Resource": "*" }  
  ]  
}
```

7. Schedule the monitoring script using **cron**.
8. Open **CloudWatch → Metrics**.
9. Verify the custom **NginxStatus** metric.
10. Create a CloudWatch alarm for **NginxStatus = 0**.
11. Configure the alarm to send notifications through the **SNS topic**.
12. Open **CloudWatch → Dashboards**.
13. Open the Server-Monitoring dashboard.
14. Add the **NginxStatus** metric to the dashboard.
15. Save the dashboard.
16. Stop Nginx: `sudo systemctl stop nginx`
17. Verify that the CloudWatch metric changes to **0**.
18. Verify that the CloudWatch alarm changes to **ALARM**.
19. Verify that an email alert is received through SNS.
20. Start Nginx again: `sudo systemctl start nginx`
21. Verify that the alarm returns to **OK**.

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#InstanceDetails:instanceId=i-055850127ddec40f1

EC2 > Instances > i-055850127ddec40f1

Instance summary for i-055850127ddec40f1 (ec2-vir-sever) Info

Updated less than a minute ago

Instance ID
i-055850127ddec40f1

Public IPv4 address
13.218.89.10 | [open address](#)

Private IPv4 addresses
172.31.21.250

Instance state
Running

Public DNS
ec2-13-218-89-10.compute-1.amazonaws.com | [open address](#)

Hostname type
IP name: ip-172-31-21-250.ec2.internal

Private IP DNS name (IPv4 only)
ip-172-31-21-250.ec2.internal

Answer private resource DNS name
IPv4 (A)

Instance type
t3.micro

Auto-assigned IP address
13.218.89.10 [Public IP]

VPC ID
vpc-0c90519b46e7b972f (vpc-virginia)

Elastic IP addresses
-

AWS Compute Optimizer finding
[Opt-in to AWS Compute Optimizer for recommendations.](#)

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```
Running transaction
Preparing
Running scriptlet:
Installing : nginxfilesystem-1:1.30.4-1.amzn2023.0.1.noarch
Installing : nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch
Installing : 11bunwind-1.4.0-5.amzn2023.0.3.x86_64
Installing : gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64
Installing : nginx-core-1:1.30.4-1.amzn2023.0.1.x86_64
Installing : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
Installing : nginx-1:1.30.4-1.amzn2023.0.1.x86_64
Running scriptlet:
Verifying : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
Verifying : gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64
Verifying : 11bunwind-1.4.0-5.amzn2023.0.3.x86_64
Verifying : nginx-1:1.30.4-1.amzn2023.0.1.x86_64
Verifying : nginx-core-1:1.30.4-1.amzn2023.0.1.x86_64
Verifying : nginxfilesystem-1:1.30.4-1.amzn2023.0.1.noarch
Verifying : nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch

Installed:
generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch      gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64      11bunwind-1.4.0-5.amzn2023.0.3.x86_64
nginx-1:1.30.4-1.amzn2023.0.1.x86_64                  nginx-core-1:1.30.4-1.amzn2023.0.1.x86_64      nginxfilesystem-1:1.30.4-1.amzn2023.0.1.noarch
nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch

Complete!
[ec2-user@ip-172-31-21-250 opt]$ sudo start nginx
sudo: start: command not found
[ec2-user@ip-172-31-21-250 opt]$ sudo systemctl start nginx
[ec2-user@ip-172-31-21-250 opt]$ sudo systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)
   Active: active (running) since Wed 2026-09-09 09:15:18 UTC; 14s ago
     Process: 35737 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
     Process: 35738 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
     Process: 35739 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
    Main PID: 35740 (nginx)
      Tasks: 3 (limit: 1059)
     Memory: 3.4M
        CPU: 59ms
    cgroup: /system.slice/nginx.service
           └─35740 "nginx: master process /usr/sbin/nginx"
             └─35741 "nginx: worker process"
               └─35742 "nginx: worker process"
```

us-east-1.console.aws.amazon.com/iam/home?region=us-east-2#/roles/details/EC2-CloudWatchAgent-Role?section=permissions

IAM > Roles > EC2-CloudWatchAgent-Role

Identity and Access Management (IAM)

Search IAM

Dashboard

Access Management

- Roles
- Policies
- IAM users
- IAM user groups
- Identity providers
- Account settings
- Root access management
- Temporary delegation requests

Access reports

Last activity
9 minutes ago

Maximum session duration
1 hour

Permissions | Trust relationships | Tags | Last Accessed | Revoke sessions

Permissions policies (2) Info

You can attach up to 20 managed policies.

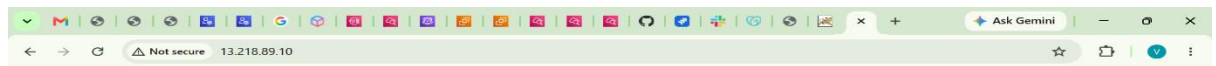
[Simulate](#) [Remove](#) [Add permissions](#)

Filter by Type

Search

☐	Policy name	Type	Att...
☐	CloudWatchAgentServerPolicy	AWS managed	1
☐	CustomCloudWatchMetricPolicy	Customer inline	0

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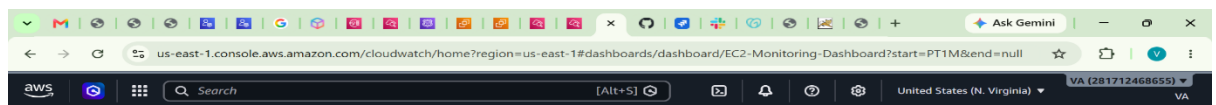


Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

Thank you for using nginx.



EC2-Monitoring-Dashb...

1

nginx

CloudShell

Agent Toolkit for AWS

Feedback

Console Mobile App

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Alarms

Mute rules

Alarms (1/1)

Clear selection

Create composite alarm

Actions

Create alarm

Filter alarms

Quick filters

< 1 >

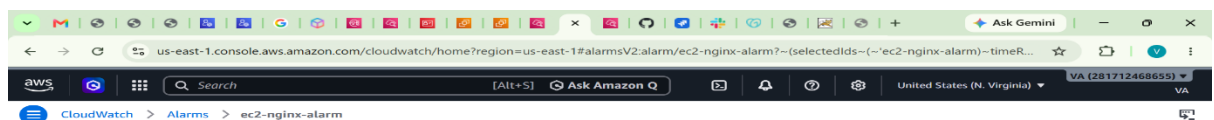
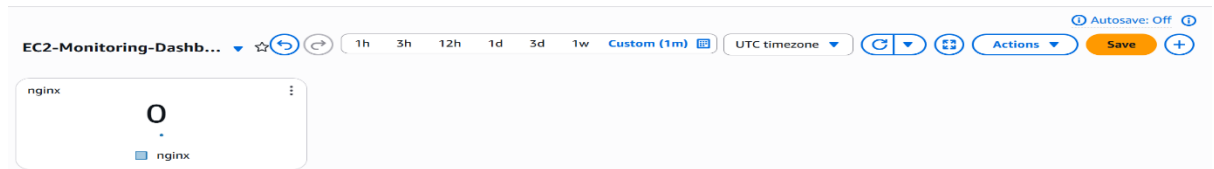
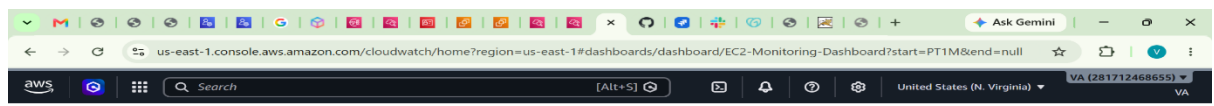
☒	Name	State	Actions	Actions muted	Last state update
☒	ec2-nginx-alarm	OK	Actions enabled	-	2026-09-09 09:09:09

```
ec2-user@ip-172-31-21-250:~$ sudo amazon-cloudwatch-agent --region us-east-1 --s3-bucket s3://cloudwatch-agent-us-east-1 --log-group /aws-logs-1-2020 (END)
Sep 09 06:05:07 ip-172-31-21-250.ec2.internal start-amazon-cloudwatch-agent[1754]: 1: Trying to detect region from ec2
Sep 09 06:05:07 ip-172-31-21-250.ec2.internal start-amazon-cloudwatch-agent[1754]: 2026/09/09 06:05:07 D: ec2tagger processor required because append_dimensions is set
Sep 09 06:05:07 ip-172-31-21-250.ec2.internal start-amazon-cloudwatch-agent[1754]: 2026/09/09 06:05:07 Configuration validation first phase succeeded
Sep 09 06:05:07 ip-172-31-21-250.ec2.internal start-amazon-cloudwatch-agent[1754]: 1: Detecting run_as_user...

[ec2-user@ip-172-31-21-250 ~]$ sudo systemctl status amazon-cloudwatch-agent
[ec2-user@ip-172-31-21-250 ~]$ sudo systemctl status nginx
nginx.service - The nginx HTTP and reverse proxy server
Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)
Active: active (running) since Wed 2026-09-09 09:15:18 UTC; 8min ago
Process: 35737 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (Code=exited, status=0/SUCCESS)
Process: 35738 ExecStartPre=/usr/sbin/nginx -t (Code=exited, status=0/SUCCESS)
Process: 35739 ExecStart=/usr/sbin/nginx (Code=exited, status=0/SUCCESS)
Main PID: 35740 (nginx)
Tasks: 3 (limit: 1059)
Memory: 3.5M
CPU: 60ms
CGroup: /system.slice/nginx.service
└─35740 "nginx: master process /usr/sbin/nginx"
    └─35741 "nginx: worker process"
        └─35742 "nginx: worker process"

Sep 09 09:15:18 ip-172-31-21-250.ec2.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...
Sep 09 09:15:18 ip-172-31-21-250.ec2.internal nginx[35738]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Sep 09 09:15:18 ip-172-31-21-250.ec2.internal nginx[35738]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Sep 09 09:15:18 ip-172-31-21-250.ec2.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
[ec2-user@ip-172-31-21-250 ~]$ sudo bash /opt/monitoring-nginx.bash
nginx is running
[ec2-user@ip-172-31-21-250 ~]$ sudo crontab -e
crontab: installing new crontab
[ec2-user@ip-172-31-21-250 ~]$ sudo crontab -l
# * * * * /opt/monitoring-nginx.bash
[ec2-user@ip-172-31-21-250 ~]$ sudo systemctl stop nginx
[ec2-user@ip-172-31-21-250 ~]$ sudo systemctl status nginx
nginx.service - The nginx HTTP and reverse proxy server
Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)
Active: inactive (dead)

Sep 09 09:15:18 ip-172-31-21-250.ec2.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...
Sep 09 09:15:18 ip-172-31-21-250.ec2.internal nginx[35738]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Sep 09 09:15:18 ip-172-31-21-250.ec2.internal nginx[35738]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Sep 09 09:15:18 ip-172-31-21-250.ec2.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server...
Sep 09 09:30:16 ip-172-31-21-250.ec2.internal systemd[1]: Stopping nginx.service - The nginx HTTP and reverse proxy server...
Sep 09 09:30:16 ip-172-31-21-250.ec2.internal systemd[1]: nginx.service: deactivated successfully.
Sep 09 09:30:16 ip-172-31-21-250.ec2.internal systemd[1]: Stopped nginx.service - The nginx HTTP and reverse proxy server.
[ec2-user@ip-172-31-21-250 ~]$
```



CloudWatch			
Favorites and recents			
Ingestion			
Dashboards			
Alarms 1 0 0 0			
AI Operations			
GenAI Observability			
Application Signals (APM)			
Infrastructure Monitoring			
Container Insights			
Database Insights			

Alarms			
ec2-nginx-alarm			
Details Tags Mute rules Actions History Parent alarms			
History (4)			
30 days up to current time			
Date (UTC)	Type	Description	
2026-09-09 09:44:27	Action	Successfully executed action arm:aws:sns:us-east-1:281712468655:EC2-Service-Alert	
2026-09-09 09:44:27	State update	Alarm updated from OK to In alarm.	
2026-09-09 09:29:27	State update	Alarm updated from Insufficient data to OK.	
2026-09-09 09:28:37	Configuration update	Alarm "ec2-nginx-alarm" created	

mail.google.com/mail/u/0/?ogbl#inbox/FMfcgzQhWLRKwWHgXVbBXrGnsVmGbwxt

Ask Gemini

Search mail

Compose

Inbox 5

Starred

Snoozed

Sent

Drafts

Purchases 1

More

Labels +

Alarm Details:

- Name: ec2-nginx-alarm
- Description:
- State Change: OK -> ALARM
- Reason for State Change: Threshold Crossed: 1 out of the last 1 datapoints [0.0 (09/09/26 09:43:00)] was less than the threshold (1.0) (minimum 1 datapoint for OK -> ALARM transition).
- Timestamp: Wednesday 09 September, 2026 09:44:27 UTC
- AWS Account: 281712468655
- Alarm Arn: arn:aws:cloudwatch:us-east-1:281712468655:alarm:ec2-nginx-alarm

Threshold:

- The alarm is in the ALARM state when the metric is LessThanThreshold 1.0 for at least 1 of the last 1 period(s) of 60 seconds.

Monitored Metric:

- MetricNamespace: nginx
- MetricName: nginx
- Dimensions: [InstanceId = i-055850127ddec40f1]
- Period: 60 seconds
- Statistic: Minimum
- Unit: not specified

Enable desktop notifications for Gmail. OK No thanks forward

32°

ENG IN

15:26 09-09-2026